

Data Mining and Applications

- ❑ Web page analysis: classification, clustering, ranking
- ❑ Collaborative analysis & recommender systems
- ❑ Basket data analysis to targeted marketing
- ❑ Biological and medical data analysis
- ❑ Data mining and software engineering
- ❑ Data mining and text analysis
- ❑ Data mining and social and information network analysis
- ❑ Built-in (invisible data mining) functions in Google, Microsoft, LinkedIn, Meta, ...
- ❑ Major dedicated data mining systems/tools
 - ❑ SAS, MS SQL-Server Analysis Manager, Oracle Data Mining Tools)



Image Source:

http://www.aia.es/wp-content/uploads/2012/09/recommendation_systems.jpg

Data Mining and Society

- ❑ Data mining technology may benefit society
 - ❑ Ex.: Help scientific discovery, business management, economy recovery, and security protection (*e.g.*, the real-time discovery of intruders and cyberattacks)
- ❑ Need to guard against the misuse of data mining
 - ❑ Data mining also poses the risk of unintentionally disclosing some confidential business or government information and disclosing an individual's personal information
- ❑ Studies on data security in data mining and privacy-preserving data publishing and data mining are important, ongoing research theme
 - ❑ The philosophy is to observe data sensitivity and preserve data security and people's privacy while performing successful data mining
- ❑ These and other related issues will be discussed throughout the book